

Networking Fundamentals for Security

Where data travels and where security matters

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A student opens the university LMS on café Wi-Fi.

The browser shows a certificate warning. Where could the problem be?

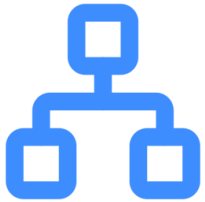


Discuss the device, the network, and the website.

By the end, you will be able to:

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1. Name the devices and follow a network path.
2. Explain IP addresses, ports, protocols, and DNS.
3. Locate a risk and choose a suitable control.



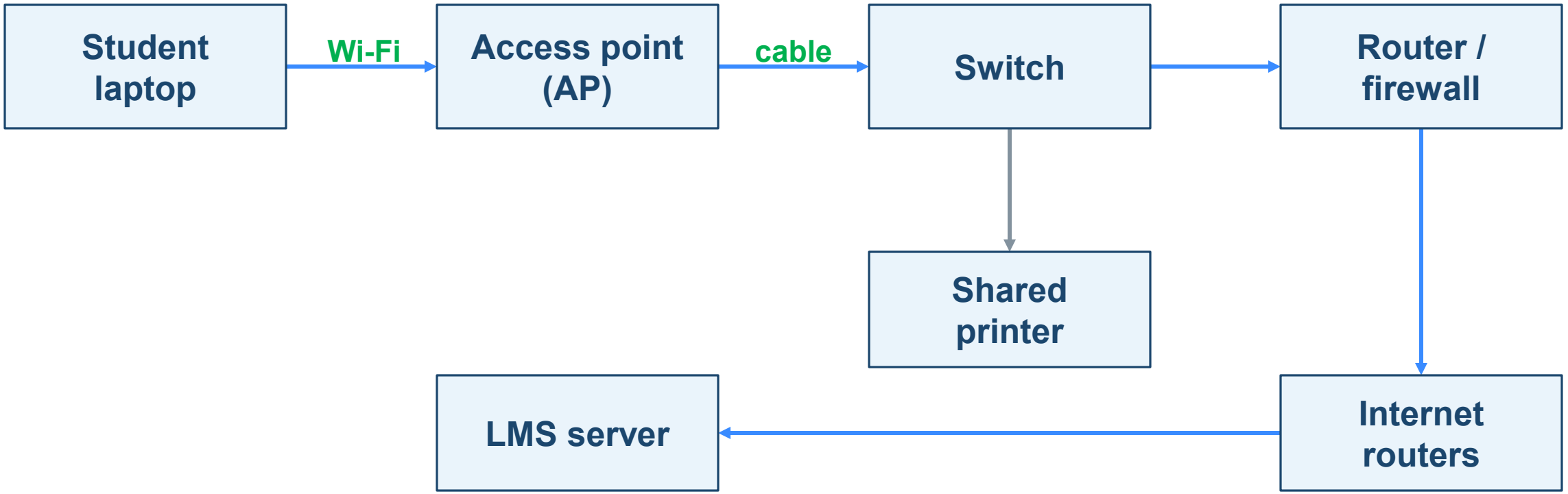
1. Follow the network path

A network connects devices so they can exchange data.

One visit to the LMS

The laptop sends a request. The server sends a response.

A SIMPLE CAMPUS LAN



Real routes can include more devices.

LAN, WAN, and the Internet

Term	Meaning	Everyday example
LAN Local Area Network	A network in a limited local area	Computers and a printer in one lab
WAN Wide Area Network	Links networks over larger distances	A university links two distant sites
Internet	Many connected networks worldwide	Your campus reaches a cloud LMS

A campus can contain many separate LANs.

Endpoints send and receive data



An endpoint is a device at an end of a communication.

Client

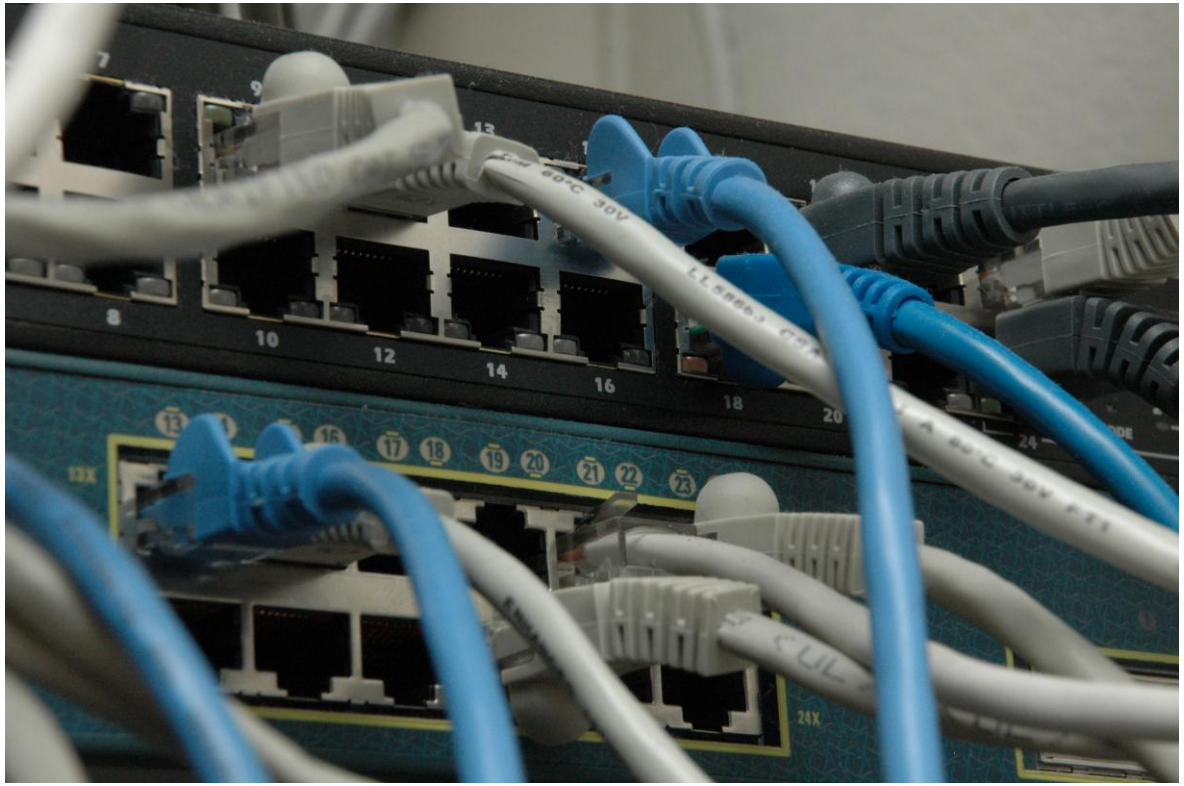
Starts a request, such as a student's browser.

Server

Provides a service, such as course files.

Printers and cameras are also connected devices.

A switch connects local devices



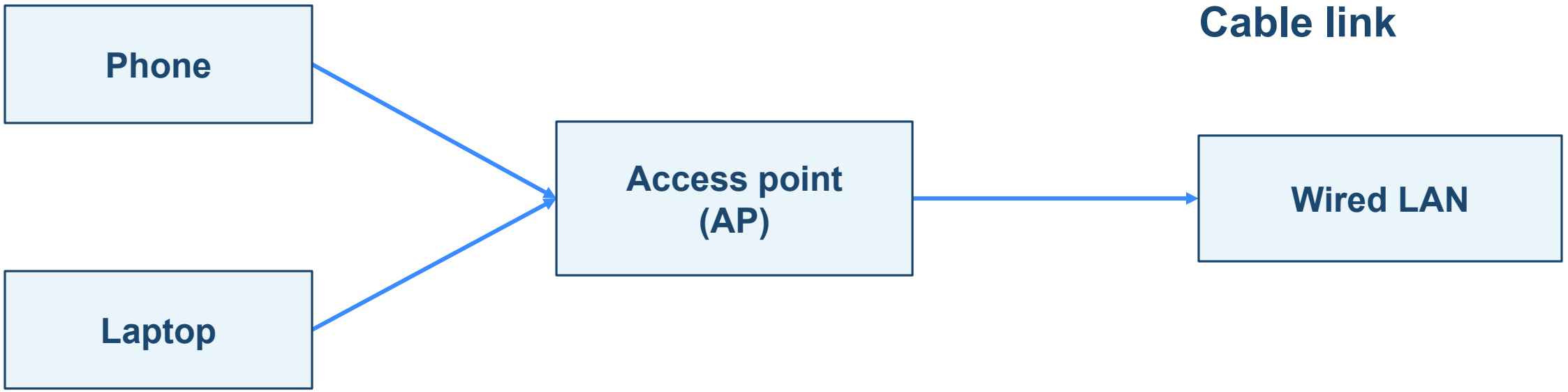
An Ethernet frame carries data across a local link.

A switch forwards Ethernet frames inside a LAN.

1. It learns local device addresses called MAC addresses.
2. An unused cable socket can be an entry point.

An access point provides Wi-Fi access

Wireless links



An AP lets wireless devices join a network.

Security concern: who can connect, and which network do they reach?

A router connects different networks



A router forwards IP packets between networks.

1. Your default gateway is the next router for other networks.
2. A home “Wi-Fi router” often combines router, AP, and switch.

Where can a network risk begin?

Location	What could go wrong?	Useful control
Device	Malware uses the device's connections	Updates and endpoint protection
Wi-Fi or cable socket	An unauthorized device joins	Network access control
Router settings	An attacker changes settings	Protected admin access and updates

A control is a measure that reduces a security risk.

Real example: WannaCry in 2017



Historical incident. The ransom screen shows the impact.

WannaCry spread to reachable, unpatched Windows systems.

Where was the weakness?

A vulnerable SMB service. SMB supports file sharing.

What could reduce spread?

Install the fix and restrict unnecessary SMB traffic.

Quick check

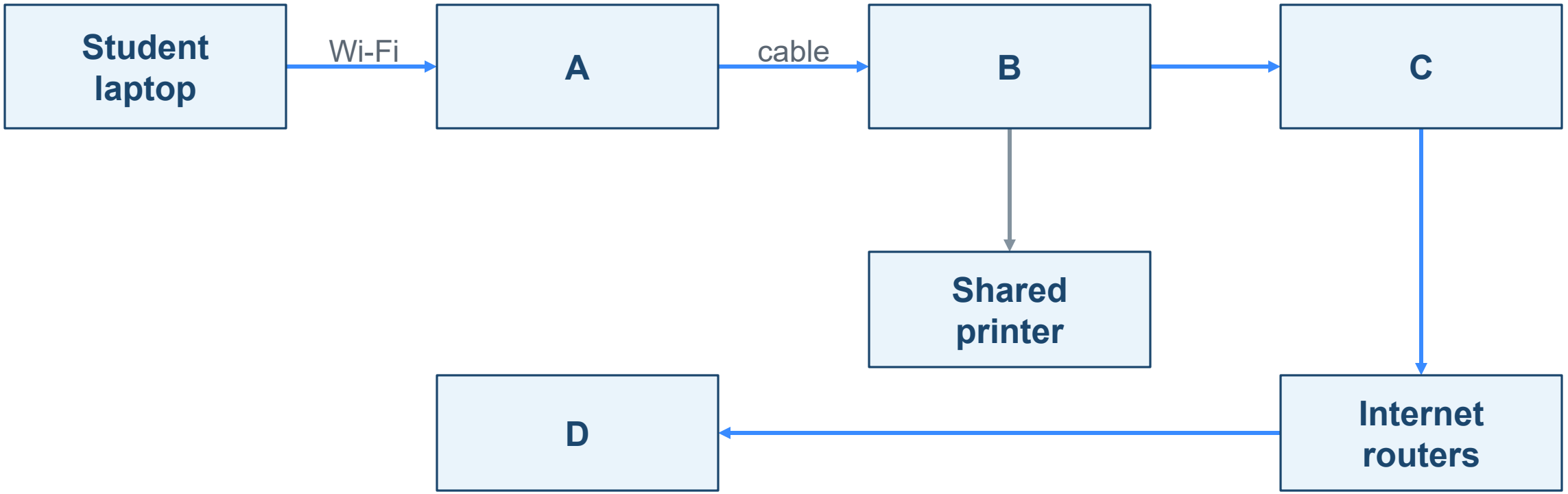
The Internet connection fails.
Can a local printer still work?



Activity: label the network

Name A–D, then trace a print job and an LMS request.

A SIMPLE CAMPUS LAN

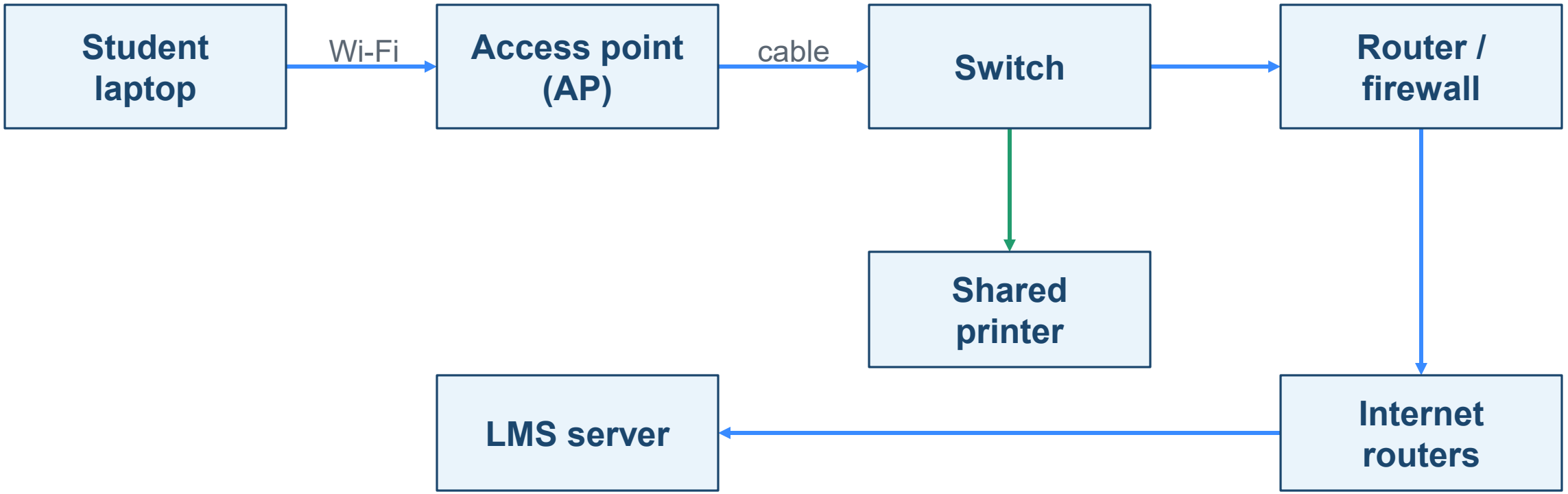


Add one risk beside a device or connection. Explain why it belongs there.

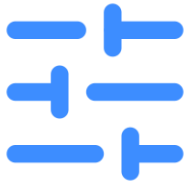
Activity answers

Printing can stay local. The remote LMS path crosses other networks.

LOCAL PRINTING PATH



Example risk: an unknown device joins Wi-Fi.
Example control: require approved network access.



2. Read a network connection

An address identifies the destination. A service follows communication rules.

An IP address helps deliver packets

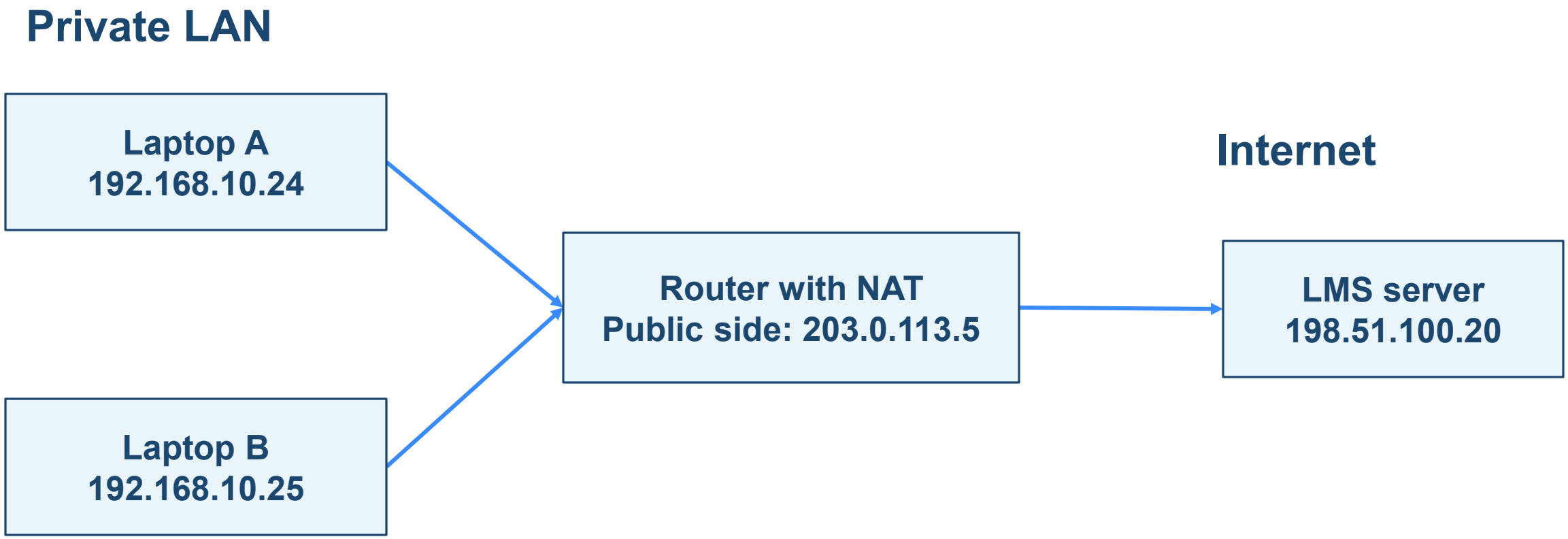
Internet Protocol (IP) address

A number for a device’s network connection (interface).

Example item	Address in our classroom model
Student laptop	192.168.10.24
Default gateway	192.168.10.1
Remote LMS server	198.51.100.20

These are example addresses. An IP address does not prove a person’s identity.

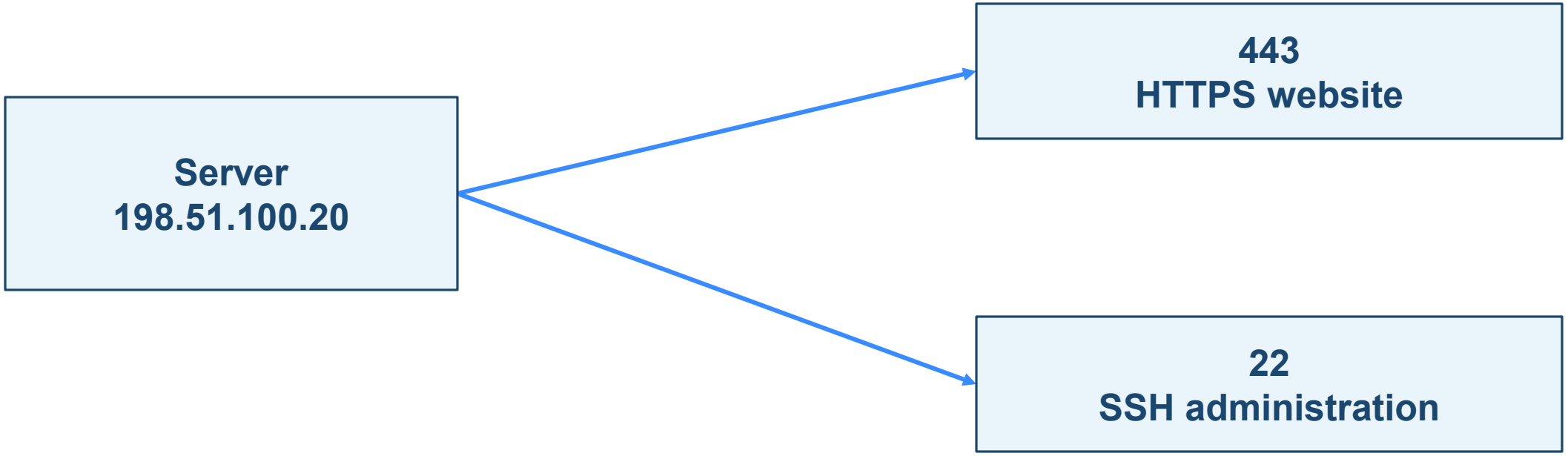
Private addresses and NAT



NAT = Network Address Translation. It changes address information as traffic passes.

A port identifies a service

One IP address can offer several services.



IP address: which network destination?
Port number: which service at that destination?

Protocols define communication rules

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Protocol	Plain English	In our web example
IP	Delivers packets between networks	Destination address
TCP	Delivers data reliably and in order	Common transport for HTTP/1.1 and HTTP/2
TLS Transport Layer Security	Protects a connection's data	Encryption and certificate checks
HTTP Hypertext Transfer Protocol	Defines web requests and responses	Request a course page

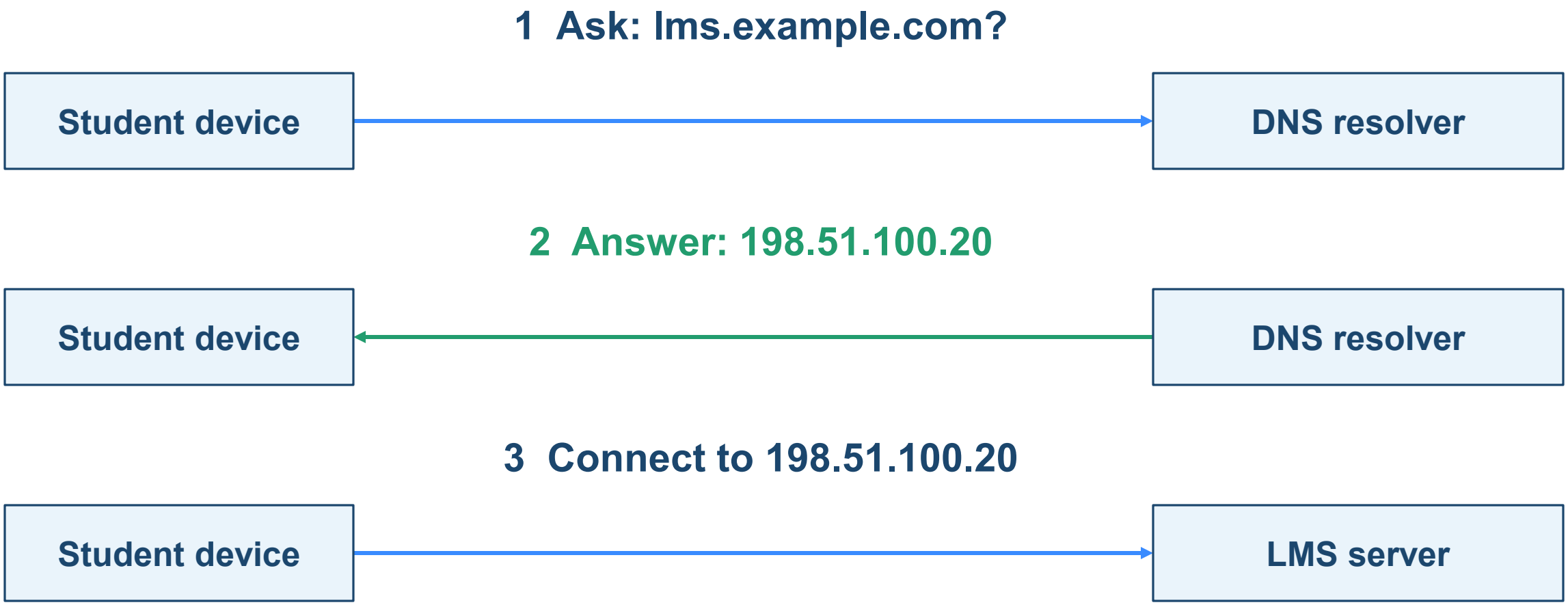
HTTPS is HTTP protected by TLS.

Certificate: digital evidence used to check the server's identity.

DNS looks up a name

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DNS = Domain Name System. A resolver finds and returns DNS information.



The URL describes a web request

https://lms.example.com/course/6

Part	Meaning
https	Use HTTP protected by TLS
lms.example.com	Domain name. DNS can find its address.
/course/6	Path to a resource on the site
Port 443	Default HTTPS port when no port is written

It is highly recommended to check the full domain name before signing in

HTTP and HTTPS on the network

HTTP	HTTPS
No TLS protection for the web content	TLS protects the web content in transit
Someone on the path may read or change it	Someone on the path cannot normally read or change it
Example readable content: course=6&student=demo	A network observer sees encrypted content when TLS works correctly

Encryption makes content unreadable without the right keys.

HTTPS has a clear protection boundary

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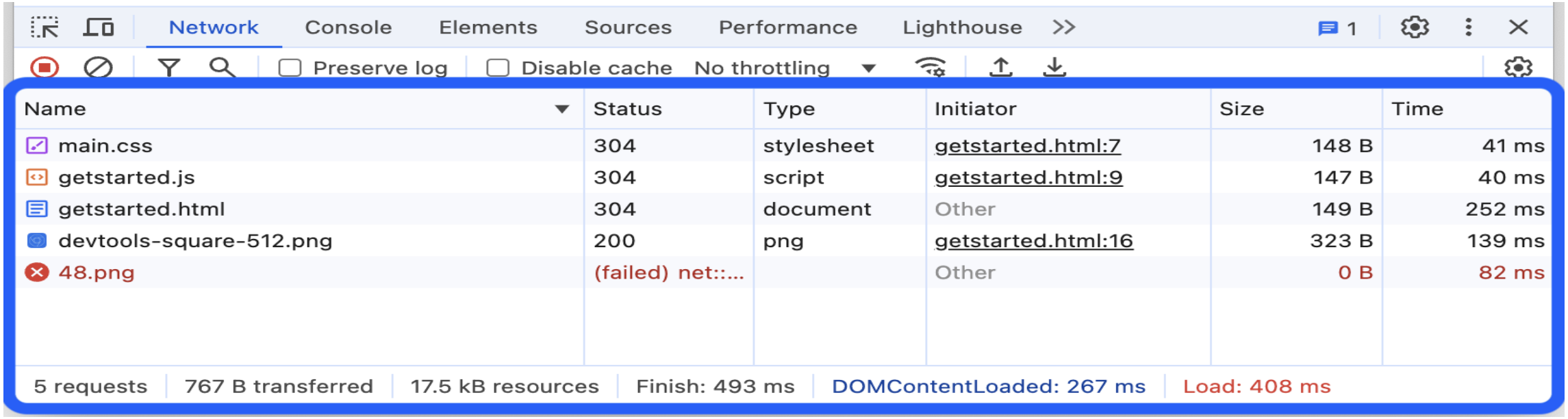
Question	What HTTPS tells us
Can a nearby observer read the page content?	Normally no, when TLS works correctly.
Is every website using HTTPS honest?	No. A phishing site can also use HTTPS.
Can malware read data on my own device?	Yes. Endpoint security still matters.
Is all network information hidden?	No. IP addresses and traffic timing may remain visible.

Demo: a page makes network requests

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https://chrome.dev/devtools-network-activity/getstarted.html

2 Open DevTools and select Network. **3 Reload the page.**

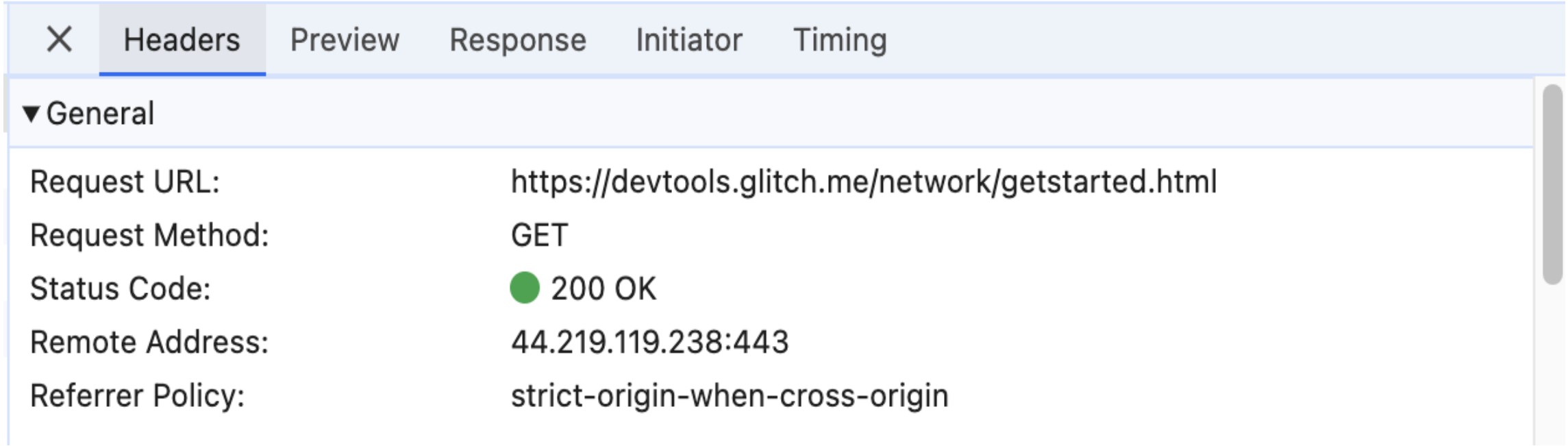


A row is a resource request, not one network packet.

One request, explained

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Click a request, then choose Headers.



Request URL = requested address. GET = request for content.

The browser can show clear content because it is one endpoint.

Quick check

Which term answers each question?



Where is it? Which service? Which rules? What address matches the name?

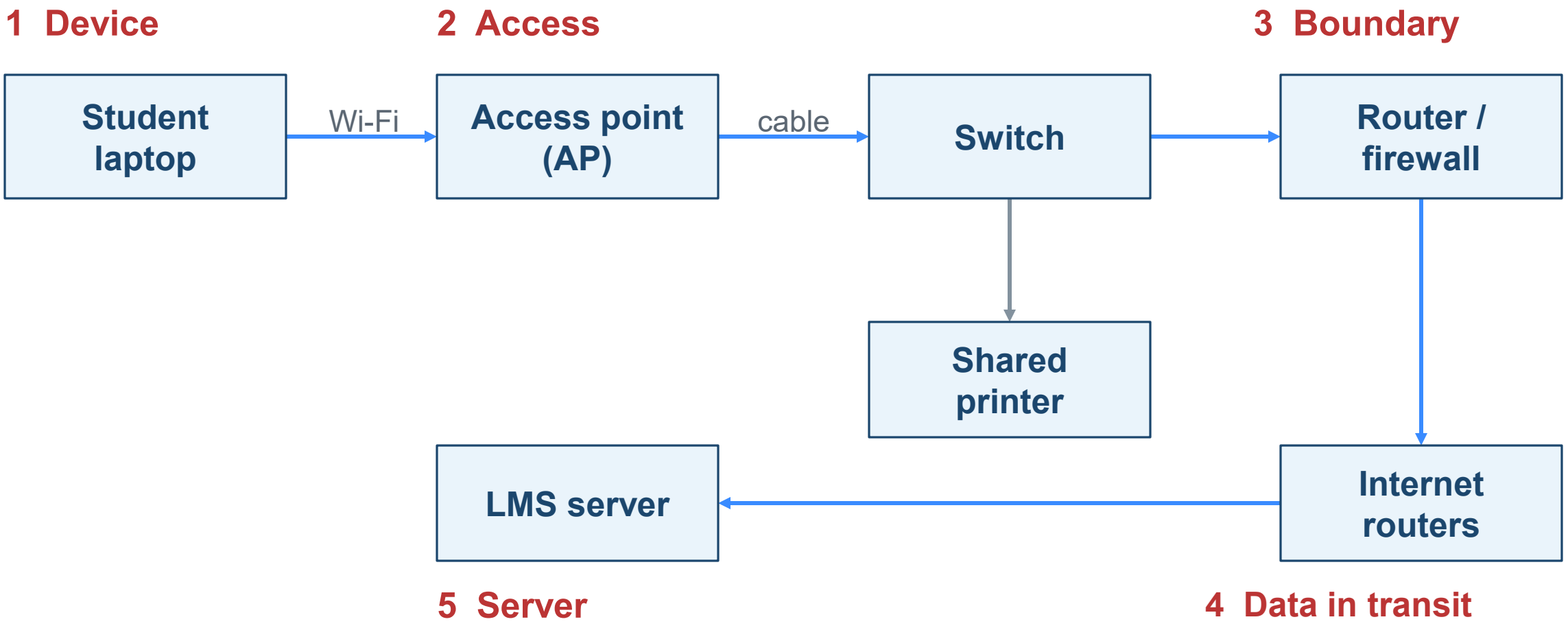


3. Locate and reduce risk

A useful explanation names the location, the weakness, and the control.

Security matters along the whole path

Point to the location before naming the control.



Where antivirus, authenticated Wi-Fi, firewall rules, TLS, and server permissions apply?

Risk at access: a fake hotspot

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Café scenario: two networks use a familiar name.

Wi-Fi name (SSID)	What the name proves
Campus_Guest	Only the advertised name
Campus_Guest	It could belong to another operator

Evil twin: a fake access point imitates a trusted Wi-Fi network.

Verify the network. Follow approved setup instructions. Stop at certificate warnings.

Risk in DNS: a false destination

DNS spoofing supplies false DNS information.

Question	Expected answer	False answer
lms.example.com?	198.51.100.20 Intended service	198.51.100.50 Wrong service

A false DNS answer can redirect traffic. It does not automatically defeat TLS.

Risk in transit: an on-path attacker

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On-path attacker: someone positioned to intercept or influence traffic.



HTTP content can be exposed.

Correct HTTPS protects content between its TLS endpoints.

Risk at a service: too much exposure

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A server should expose only the access it needs.

Connection	Example rule	Reason
Public users to HTTPS 443	Allow required website traffic	Students need the LMS
Public users to SSH 22	Deny public administration	Limit admin access
Approved admin path to SSH	Allow authorized admins	Maintain the server securely

Firewall: software or hardware that allows or blocks traffic using rules.

Risk at the server: excessive access



The LMS receives an encrypted request.
It must still decide what the user may read.

Authentication

Who are you? The system checks your identity.

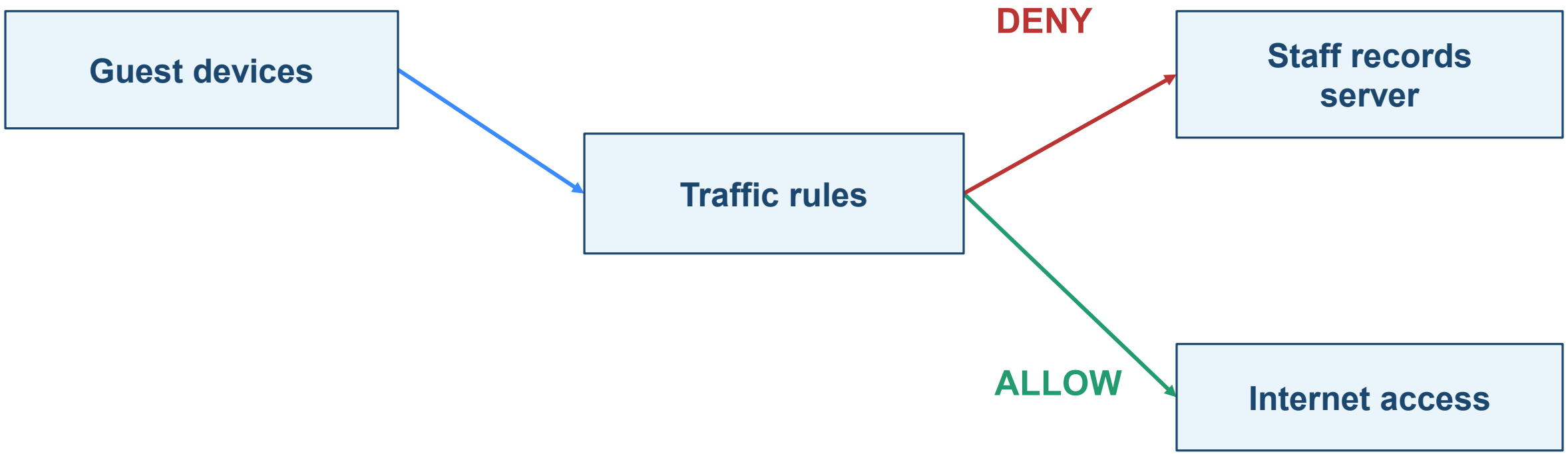
Authorization

What may you do? The system checks your permissions.

Segmentation limits unnecessary paths

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Segmentation divides a network into groups and controls communication between them.



A risk explanation needs a location

Location	Weakness or concern	Control that fits
Laptop	Unpatched software	Install supported updates
Wi-Fi access	Unknown network operator	Verify the approved connection
Web traffic	Sensitive data over HTTP	Use valid HTTPS
Private service	Access from unnecessary users	Limit reachability and permissions

Activity: a small office network

A small office shares one network.

1. Guests connect to Wi-Fi for Internet access.
2. Staff use a payroll server. A printer runs old firmware.
3. The server allows administration from any Internet address.

Draw the path, mark two risks, and place two controls.
Explain how each control helps.

Activity: one defensible answer

Risk	Control	Why it helps
Guests can reach staff systems	Separate guest access and enforce traffic rules	Limits access to private services
Administration is public	Allow only an approved admin path	Reduces unnecessary exposure
Printer firmware is old	Update it, or restrict it while arranging replacement	Reduces exploitable weaknesses

Q&A: questions 1–5

01 What is the difference between a LAN and a WAN?

02 What does an access point do?

03 How do a switch and a router differ?

04 What does an IP address identify?

05 What does a port number identify?

Q&A: answers 1–5

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1. A LAN covers a local area. A WAN links networks over larger distances.
2. An AP lets wireless devices join a network.
3. A basic switch forwards within a LAN. A router forwards between networks.
4. An IP address identifies a network interface for packet delivery.
5. A logical port identifies an application or service endpoint.

Q&A: questions 6–10

06 What does DNS do when you open a website?

07 What protection does HTTPS add?

08 Can a phishing website use HTTPS?

09 Why does an exposed service need attention?

10 How can segmentation reduce harm?

Q&A: answers 6–10

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6 DNS can return an address for the website’s domain name.

7 TLS protects web content in transit and checks the server certificate.

8 Yes. HTTPS alone does not prove the site’s owner is honest.

9 A reachable service may allow access or expose a weakness. Limit and maintain it.

10 It limits unnecessary communication and can restrict an incident’s spread.

Sources and further reading

Network devices	Cisco: Switch vs router
DNS	Cloudflare: What is DNS?
HTTP and HTTPS	Cloudflare: What is HTTPS?
Real incident	Microsoft: WannaCrypt, 12 May 2017
Browser demonstration	Google: Inspect network activity
Network separation	Cisco: Network segmentation

Links are clickable. Image credits and teaching notes appear in speaker notes.

Assignment: your own network path

Choose one: opening a course page, making a video call, or using a shared printer.

1. Draw the endpoints and the network devices in between.
2. Explain where IP addresses, ports, or DNS help in your example.
3. Mark two possible risks and add a suitable control for each.

Submit one PDF to LMS: diagram + short explanations.

Five ideas to keep after today

- Wi-Fi connects you to a network. It is different from the Internet.
- Switches connect local devices. Routers connect networks.
- IP addresses, ports, and protocols answer different questions.
- DNS finds an address. HTTPS protects web content in transit.
- Security controls are needed at the device, on the path, and at the service.

Next week: network defense and secure Wi-Fi